

Report for ADL project
Motorized Automatic Pitcher For Guitar

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Chapter 1

Introduction

This project is realized completely in PL (Programmable Logic).

1.1 Sampling

The sampling of the data is performed using the ADAU1761 chip of the Zed-Board. It is a coder/decoder that samples the voltage level coming from the line-in, and can also output it on the line-out and do all sorts of transformations. The process of sampling has been facilitated by the use of the interface developed by Mike Field [1].

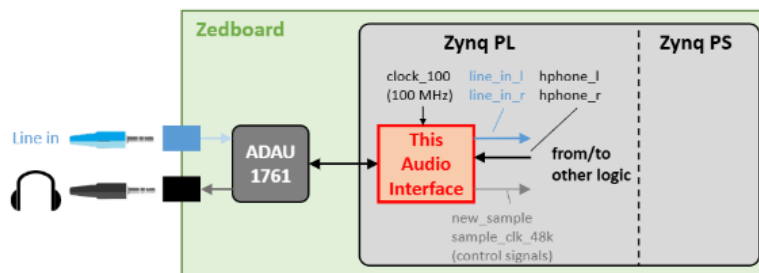


Figure 1.1: This is a figure

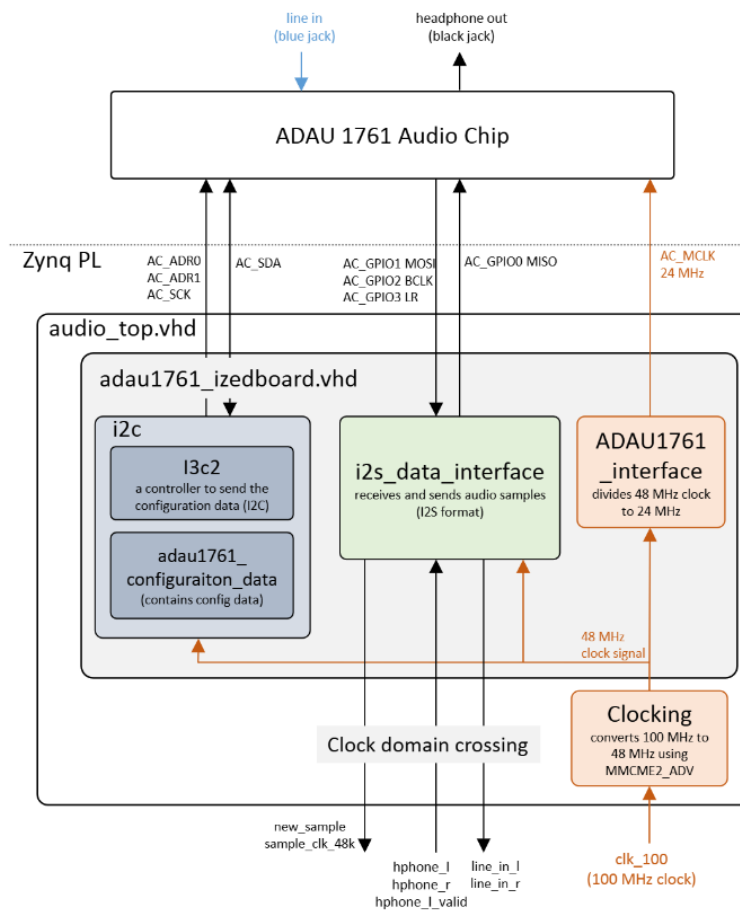


Figure 1.2: Audio interface block diagram.

1.2 Downsampling

We aim to perform the FFT on our samples to detect what is the frequency with the maximum power. But what resolution can we achieve? What is the range of frequencies we can measure?

When we fill the FFT buffer, we are capturing a specific “window” of time. If we collect N samples, and take a new sample f_s times per second, the total time it takes to fill that buffer is:

$$T = \frac{N}{f_s} \quad (1.1)$$

The FFT assumes that the chunk of data we captured is one repeating cycle. Therefore, the absolute lowest frequency the FFT can see (other than 0 Hz DC) is a wave that completes exactly one full cycle inside our time window T . Thus:

$$f_{\min} = \frac{1}{T} \quad (1.2)$$

Substituting Equation 1.1 into Equation 1.2, we get:

$$f_{\min} = \frac{f_s}{N} = \Delta f \quad (1.3)$$

Since every bin of the FFT is spaced between one another in multiples of Δf .

We derived the equation that tells us the resolution AND the minimum frequency detectable, but what about the maximum? The Nyquist theorem tells us that the maximum frequency we can detect through sampling is:

$$f_s > 2 \cdot f_{\max} \quad (1.4)$$

So with N fixed, to get a finer resolution (smaller Δf), we want to **lower** f_s , but not too much; otherwise, we will not be able to detect some needed frequencies according to Equation 1.4.

Here is where **downsampling** comes into play. We can decide how many samples to keep to achieve a certain resolution. But instead of discarding 9 samples out of 10, we accumulate those 10 values into a register and use that as our downsampled value. This method prevents spikes given by physical events (like picking a string) by effectively computing an *unscaled average*. (Since we are looking for the biggest value in the FFT, we do not need to scale it back down).

It also helps with another problem, **aliasing**: the Nyquist theorem (Equation 1.4), also tells us that if the guitar produces some high frequency noises (which it does), the low f_s will produce some low-frequency noise into the elaborated data. This accumulation helps with that because it acts as a **low pass filter**:

In signal processing, summing N subsequent samples is the same as **convolving** your audio with a "rectangular window", which in the frequency domain it has a frequency response given by the *sinc* function:

$$H(f) = \frac{\sin(\pi f M T_s)}{\sin(\pi f T_s)} \quad (1.5)$$

Where M is the downsample factor (10), and T_s is the sample period.

The low-frequencies are going to get amplified, while the high frequencies, which complete one or multiple cycles during the window, will have their positive and negative halves cancelled out.

1.3 The ping pong buffer

To prevent data loss while the FFT component is elaborating the data, we set up two buffers of 8192 entries in BRAM, and when one is full we start the FFT on that data and start pushing new samples on the other buffer.

1.4 The FFT IP

For performing the FFT we use the Xilinx component set up like that: The output of the FFT are the frequencies components values spread across 8192 bins.

1.5 Smart peak finder

To find the fundamental frequency we want to find the most powerful component. First problem is that the resulting 8192 bins, having the sample rate at 4800 hz, are:

- Bins 0 to 4095 represent the true positive frequencies from 0 Hz to 2,400 Hz (Nyquist Limit).

- Bins 4096 to 8191 represent the negative frequencies wrapping backwards from 4,800 Hz down to 2,400 Hz.

Chapter 2

Methods

Chapter 3

Results

Bibliography

- [1] Mike Field. zedboard audio interface. https://github.com/ems-kl/zedboard_audio, 2015.